DOES MEMORY PROVENANCE MATTER? MATCHED SUCCESS–FAILURE SWAPS LEAVE THE CAUSAL SIGN UNRESOLVED

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ABSTRACT

Self-evolving agents convert previous trajectories into reusable textual memory. We study a variable carried into that state that content-only audits can miss: the process provenance of the trajectory from which memory was written. A recent financial-agent audit reports 0.931 accuracy over 101 success-derived ReasoningBank retrievals versus 0.647 over 34 failure-derived retrievals, with all ten reported persistent W→W failures in the failure-derived stratum. We introduce a provenance-audit identification ladder that separates observational provenance signals, controlled writer-mode contrasts, exact-information metadata interventions, and source-faithful transport. On released WebArena evidence, R4 evaluates terminal success over four task pairs that pass a frozen pre-outcome semantic-equivalence rule and 48 completed requests. The observed success-derived minus failure-derived difference is +0.1667, but the frozen permutation test gives $p = 0.0785$; with four independent tasks, the exact one-sided sign-flip floor is $1/16 = 0.0625$, so the endpoint is structurally non-decisive under the frozen 0.05 rule. R6 uses 23 disjoint task pairs and 276 completed requests. Its observed reference-action-match difference is -0.0942 with counterevidence-side $p = 0.0792$; paired next actions differ in 21.0% of seeds, but this readout is not terminal quality. The two L1 endpoints are therefore not pooled: their descriptive directions disagree across distinct estimands, so the current evidence does not support an endpoint-invariant causal sign. R5 targets the cleaner exact-information rung by holding actionable memory byte-identical and changing only provenance metadata, but its frozen ten-task design has only five eligible unique tasks and stops before any model call. Thus the strong observational association survives as an audit signal while the provenance-only causal sign remains unresolved. The reusable contribution is an explicit identification protocol that reports the intervention channel, independent inference unit, assumptions, and strongest allowed claim before provenance is used in memory trust decisions.

1 INTRODUCTION

Self-evolving agents increasingly learn through external memory. A successful episode can become a reusable workflow; a failed episode can become a corrective reflection. Systems such as ReasoningBank convert trajectories into reasoning memories that influence future behavior (Ouyang et al., 2026), while Agent Workflow Memory and Reflexion persist workflows or verbal feedback across trials (Wang et al., 2024; Shinn et al., 2023).

The usual abstraction treats retrieved content as the state variable that matters. Content-centric systems such as A-MEM and MemRL improve memory organization, retrieval, or runtime utility (Xu et al., 2025; Zhang et al., 2026). A second literature makes provenance explicit: recent work organizes execution provenance and evidence tracing for LLM agents (Wang et al., 2026), while Memory Provenance Laundering shows that consolidation can obscure low-trust source provenance while preserving an action trigger (Xu et al., 2026). We therefore do not claim that provenance-aware memory is itself new. Our narrower object is *process provenance*: whether persistent guidance was produced from a successful or failed trajectory, and which causal channel is actually changed when that provenance is manipulated.

Failure provenance also has no justified negative sign a priori. Learning from Failure converts failed computer-use trajectories into diagnoses and patches that improve later behavior (Sun et al., 2026), and controlled multi-round skill-evolution experiments report useful skills in settings that include failure feedback (Liu et al., 2026). These results rule out the shortcut "failure-derived means harmful." A valid audit must separate an observational provenance association from the intervention used to test it.

A recent audit of self-evolving financial agents supplies the motivating signal (Li & Zhu, 2026). In its ReasoningBank analysis, success-derived retrievals have reported accuracy 0.931 over 101 cases and failure-derived retrievals 0.647 over 34 cases. The audit also reports ten persistent W→W failures, all in the failure-derived retrieval stratum. This is a strong association, but task difficulty is an obvious confound: hard tasks can be more likely to fail during acquisition and remain difficult later. A second ambiguity is intervention design. Switching ReasoningBank's success- versus failure-reflection objective changes an outcome-conditioned writer; it is not the same operation as holding memory text fixed and changing only provenance metadata.

We formalize these distinctions as a *provenance-audit identification ladder*. In plain terms, L0 only compares provenance-labeled groups; L1 rewrites memory under different success/failure reflection objectives; L2 keeps actionable memory text fixed and flips only visible provenance metadata; L3 repeats an identified intervention in the original source runtime. Each rung names the intervention, independent inference unit, identifying assumptions, and strongest allowed conclusion. This makes the protocol applicable beyond ReasoningBank to memory systems in which history changes a writer, a retrieval rule, or explicit metadata.

Our controlled evidence populates the ladder without manufacturing a causal sign. R4 instantiates L1 with four independent terminal task units and 48 completed requests: the observed success-derived minus failure-derived terminal-success difference is +0.1667, but $p = 0.0785$ and four units make a one-sided exact $p < 0.05$ unattainable. R6 supplies 23 disjoint L1 task units and 276 completed requests: paired actions differ in 21.0% of seeds, while the observed reference-action-match difference is -0.0942 with counterevidence-side $p = 0.0792$; reference-action match is not terminal quality. Because the endpoints are different estimands and their descriptive directions disagree, the current L1 evidence supports no endpoint-robust directional claim. R5 targets L2 with byte-identical actionable memory, but the frozen ten-task design exposes only five eligible unique tasks and stops before any model call. Source-faithful L3 replication remains support debt. C4 therefore remains active without a supported provenance-only causal sign; C5 remains a design implication rather than a demonstrated governance benefit.

Contributions and closest-work boundary. We contribute (i) a source-bound process-provenance association from the financial-agent audit, (ii) an L0–L3 audit contract that binds each claim to a concrete intervention channel and independent unit, and (iii) executed L1 endpoints plus an L2 support stop that make the unresolved boundary explicit instead of collapsing all evidence into one "provenance effect." The novelty claim is not new potential-outcomes algebra: it is the agent-memory audit object, its constructive non-upgrade rule (L1 cannot become L2 without additional information identity), and a reporting protocol that remains valid when experiments stop for power or support reasons. We do not claim the first provenance-aware memory system, that failure-derived memories are generally harmful, or that provenance gating improves downstream utility.

2 Provenance Audit and Identification Targets

Consider an acquisition episode with task x, trajectory τ, and terminal outcome $z \in \{0, 1\}$. Let $P \in \{S, F\}$ denote whether the source trajectory succeeded or failed. In a provenance-conditioned memory system, P can affect at least two downstream objects: the writer mode $W(P)$ used to construct memory and, when exposed to the agent, a provenance label or metadata field $L(P)$. The written memory is

$$M = G(\tau, W(P)), \tag{1}$$

and future behavior follows

$$Y = H(x', M, L, S), \tag{2}$$

where x' is the future query and S is the remaining agent state. ReasoningBank instantiates the writer channel directly: successful trajectories are analyzed under a success-reflection objective, while failed

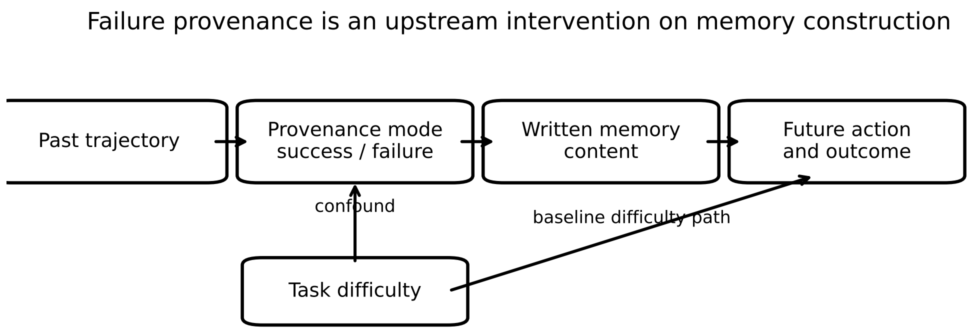

Figure 1: Coarse causal structure of provenance-conditioned memory. Acquisition difficulty can affect both source outcome and later task performance. Writer mode mediates the path from source provenance to written memory; an optional visible provenance-metadata channel is omitted for readability. Matching future queries and semantic guidance reduces the task-difficulty confound but does not make generated memories identical.

trajectories are processed under a failure-reflection objective (Ouyang et al., 2026). A system may also attach an explicit provenance field to the resulting memory.

2.1 A PROVENANCE EFFECT IS NOT A SINGLE ESTIMAND

The observational quantity

$$A_P = \mathbb{E}[Y \mid P = S] - \mathbb{E}[Y \mid P = F] \tag{3}$$

is useful as an audit signal, but it is not causal: acquisition difficulty D can influence both P and future outcomes. More importantly, a notation such as $do(P)$ is underspecified once provenance controls memory construction. One must state which downstream channel is intervened on.

The *writer-mode contrast* changes the reflection objective while fixing the future query and released observation state:

$$\Delta_W(x') = \mathbb{E}[Y \mid do(W = W_F), x', E, \mathcal{G} = 1] - \mathbb{E}[Y \mid do(W = W_S), x', E, \mathcal{G} = 1], \tag{4}$$

where E denotes the frozen future evidence/state and $\mathcal{G}$ is a preregistered information-equivalence gate on the two generated memories. This is the object approximated by our R4 and R6 bridges. Even when $\mathcal{G} = 1$, the intervention can change wording, emphasis, decomposition, or other decision-relevant properties of M; therefore Δ_W is a writer-mode-bundle estimand, not automatically a provenance-metadata-only effect.

A stricter *exact-information metadata contrast* instead holds the actionable memory text m byte-identical and changes only visible provenance metadata:

$$\Delta_L(m, x') = \mathbb{E}[Y \mid M = m, do(L = F), x', E] - \mathbb{E}[Y \mid M = m, do(L = S), x', E]. \tag{5}$$

This is the cleanest provenance-only object when the deployment system exposes provenance metadata to the policy. Our preregistered R5 design targets this rung, but its support gate fails before model execution because the frozen cohort contract supplies only five unique tasks for a ten-task target.

2.2 PROVENANCE-AUDIT IDENTIFICATION LADDER

Table 1 turns these distinctions into a reusable audit protocol. The key reporting rule is that request count is not automatically the inference unit: R4 contains 48 completed requests but only four independent task units; R6 contains 276 requests but 23 independent task units.

Table 1: Provenance-audit identification ladder.

Rung	Intervention / hold fixed	Current evidence and independent unit	Strongest allowed claim
L0	Observe provenance strata; no intervention	Financial audit; source/retrieval cases	Association / audit signal only
L1	Switch writer mode; fix future query/state; require $\mathcal{G} = 1$	R4: 4 task pairs; R6: 23 task pairs	Writer-mode-bundle contrast, not provenance-only
L2	Fix actionable memory byte-identical; switch visible metadata	R5: target 10 tasks, capacity 5, 0 calls	Provenance-metadata effect if support gate passes
L3	Repeat L1/L2 in original model/runtime/source artifacts	Financial AgentDojo replication debt	Transport evidence only

Table 2: Provenance-conditioned ReasoningBank outcomes from the released financial-agent audit.

Retrieved memory provenance	Cases	Correct	Accuracy
Success-derived	101	94	0.931
Failure-derived	34	22	0.647

2.3 Decision rule and prospective resolution

The frozen $|\Delta| \geq 0.15$ effect floor and task-level $p < 0.05$ gate are distinct. R4's four tasks imply minimum exact one-sided $p = 1/16$; R6 has 23 tasks but misses both magnitude and significance. Future confirmatory L1 task count must be prospectively powered before outcomes (Appendix A.9); no retrospective power claim upgrades the present evidence.

2.4 Identification statements

Let $Y_W(w) = H(x', G(\tau, w), L_0, S)$ denote the potential outcome under writer mode w with metadata fixed at L_0, and let $Y_L(\ell; m) = H(x', m, \ell, S)$ denote the potential outcome under metadata ℓ with actionable memory fixed exactly at m.

Proposition 1 (rung validity). (i) L0 does not identify either Δ_W or Δ_L without an ignorability assumption for source provenance, because a latent acquisition difficulty can jointly affect P and Y. (ii) Within a preregistered L1 admitted set, if writer mode is the only manipulated channel, future state E is fixed, and consistency/no-interference hold, the paired task-level contrast identifies the finite-set writer-mode effect $\mathbb{E}[Y_W(W_F) - Y_W(W_S)]$ for that admitted set. (iii) L1 does not identify Δ_L when $G(\tau, W_F) \neq G(\tau, W_S)$. (iv) Under L2, if $M = m$ is byte-identical and metadata is the only manipulated channel, the paired task-level contrast identifies $\mathbb{E}[Y_L(F; m) - Y_L(S; m)]$ for the evaluated substrate. L3 changes the target domain, not the intervention channel, so transport assumptions are additional to (ii)–(iv).

The non-upgrade in (iii) is constructive: two response functions can agree on every L1-observed point (M_S, L_0) and (M_F, L_0) yet differ arbitrarily on the unobserved metadata counterfactuals (m, L_S) and (m, L_F). Thus no amount of L1 replication alone determines an L2 effect. For our L1 gate, malformed memories, answer leakage, insufficient embedding similarity, verifier-detected actionable asymmetry, or a unique task-solving fact are rejected before outcomes; this narrows the writer-mode bundle but does not supply the missing equality assumption. Consequently a strong L0 signal cannot be promoted to L1/L2, a non-decisive L1 result cannot rescue L2, and an L2 support failure is not counterevidence. The rung and independent unit are therefore part of the claim, not bookkeeping.

3 Released Evidence for a Provenance Effect

We begin from the public aggregate audit of ReasoningBank on benign AgentDojo Banking endpoints (Li & Zhu, 2026). The audit separates retrievals according to whether the retrieved memory was derived from a successful episode or from a failed episode. Table 2 reproduces the reported counts after converting the rounded accuracies back to their integer support.

The observed accuracy gap is 28.4 percentage points. This is a large effect at the retrieval stratum level. The transition audit gives the provenance signal a sharper interpretation. ReasoningBank

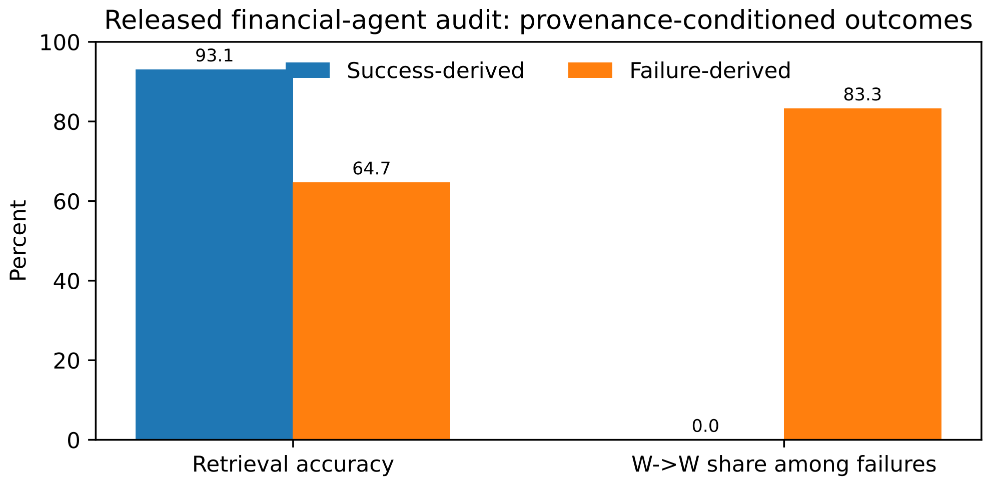

Figure 2: The released audit shows a large accuracy gap by memory provenance. Persistent W→W failures are concentrated in failure-derived retrievals.

records 25 W→C recoveries and 9 C→W regressions. Ten W→W persistent failures remain after evolution. All ten belong to the failure-derived retrieval stratum.

The rounded provenance accuracies imply 12 failures among 34 failure-derived retrievals and 7 failures among 101 success-derived retrievals. The ten W→W cases therefore account for roughly 83.3% of failures in the failure-derived stratum. The success-derived stratum contains no W→W case in the audit. Figure 2 visualizes this decomposition.

3.1 The signal is strongest for persistence

The transition structure changes the scientific interpretation of the aggregate accuracy gap. Failure-derived memory is tightly associated with errors that persist across self-evolution. The source audit identifies task difficulty as a confound for this association. Our causal formulation preserves the persistence signal as the main target and removes task difficulty through an intervention on memory provenance.

A simple arithmetic check also constrains the mechanism. The failure-derived stratum contains roughly two failures beyond the ten W→W cases. The success-derived stratum contains seven failures. The provenance signal therefore concentrates on persistent error much more strongly than on newly induced regression. This pattern motivates a paper claim centered on failure persistence.

3.2 Independent mechanism evidence

Memory Reward Inflation studies a different memory architecture and identifies an amplification channel for erroneous memories (Asadolahi et al., 2026). Incorrect memories can receive inflated utility scores, which increases their future reuse. The paper also shows that this behavior can persist under similarity-based retrieval. This mechanism is compatible with the provenance effect studied here. A failure-derived memory can carry corrective advice while remaining anchored to a trajectory whose process already demonstrated an error mode. Repeated reuse can then sustain that mode when the corrective abstraction is incomplete.

Together, the financial-agent audit and reward-inflation results motivate a causal model in which memory provenance can change the distribution of future behavior. Section 4 tests the directional provenance hypothesis with two frozen controlled endpoints.

Table 3: L1 terminal-success writer-mode contrast. Δ is success-derived minus failure-derived terminal success.

Released task	Success-derived	Failure-derived	Δ
167	0.000	0.000	0.000
23	1.000	0.667	+0.333
387	0.000	0.000	0.000
388	0.500	0.167	+0.333
Mean	0.375	0.208	+0.167

4 CONTROLLED WRITER-MODE BRIDGE

We execute matched outcome-conditioned writer-mode interventions on released WebArena evidence using the first-party ReasoningBank memory-writing contract. The financial AgentDojo audit supplies the motivating association. Within each downstream pair we fix the future query and observation evidence and apply a preregistered information-equivalence gate to the generated memories. Because the success and failure branches use different reflection objectives, this bridge tests the writer-mode bundle associated with provenance rather than changing provenance metadata alone.

4.1 TERMINAL-SUCCESS ENDPOINT

We freeze six candidate future tasks before memory writing or downstream evaluation. For each task, the writer receives paired acquisition traces under its success-reflection objective and its failure-reflection objective. The downstream query evidence remains fixed within each pair.

A pair enters terminal evaluation only after an information-equivalence gate. Both memories must parse as valid ReasoningBank items. Their embedding cosine must be at least 0.75. An independent verifier must judge the actionable guidance equivalent and find no task-solving fact unique to either memory. Literal benchmark-reference leakage excludes a pair before downstream outcomes. Four of the six frozen candidates pass, with cosines 0.902–0.931. Re-thresholding the archived pre-outcome values at 0.85 or 0.90 retains all 4/4 pairs. However, a previously archived post-hoc audit with stronger semantic reviewers is less permissive (0/4 fully equivalent under one reviewer and 1/4 under another), so we do not interpret cosine stability as proof of information identity.

The evidence packets contain repeated global browser chrome. A condition-blind projection removes complete menu blocks before the clean run. It also drops blank lines and exact duplicate lines. The projection is frozen without consulting downstream answers. A context preflight shows that the largest prompt occupies 5,209 input tokens inside a 12,288-token allocation. Qwen2.5-32B remains the executor at temperature 0.6. The pair set and seed schedule remain frozen.

Each qualified pair receives six paired seeds per writer condition. Terminal correctness uses the WebArena must-include evaluator. Table 3 reports pair-level success rates.

The mean controlled difference is +0.1667. The preregistered support gate requires a difference of at least 0.15 together with a one-sided stratified permutation value below 0.05. The magnitude criterion is met. The 100,000-permutation test gives $p = 0.0785$, so the support gate remains closed. More fundamentally, four independent task units permit only $2^4 = 16$ exact sign patterns; the smallest attainable exact one-sided value is therefore $1/16 = 0.0625$. R4 is structurally incapable of clearing the frozen 0.05 gate and is treated as directional, non-decisive evidence. A deterministic lexical audit also shows that writer mode changes pragmatic register: across the six R4 pairs, failure-derived memories use more search/filter and strong-directive language in 6/6 pairs. This reinforces the L1 writer-bundle interpretation rather than a provenance-only interpretation.

4.2 DISJOINT EARLY-ACTION ENDPOINT

The manuscript specifies action divergence before terminal feedback as a second endpoint. We execute that endpoint on a disjoint cohort after the terminal run. Candidate selection occurs before counterfactual memory writing. A released successful trajectory supplies a frozen reference action at an intermediate browser state. This reference is external to the L1 writer-mode contrast.

Table 4: Two frozen L1 writer-mode endpoints. Support requires $\Delta \geq 0.15$ with $p_S < 0.05$. Counterevidence requires $\Delta \leq -0.15$ with $p_F < 0.05$.

Endpoint	Units	Requests	Δ	Relevant p	Gate
Terminal success	4	48	+0.167	$p_S = 0.0785$	closed
Reference-action match	23	276	-0.094	$p_F = 0.0792$	closed

The candidate rule selects the earliest visible single-click decision at step two or later. Prior D2 task IDs are excluded. The remaining tasks are ordered by numeric task ID. The first 24 candidates are frozen. Their compact process traces are byte-identical across writer conditions. Terminal success labels are removed from those traces before the ReasoningBank success-reflection and failure-reflection interventions.

Twenty-three of the 24 candidates pass the pre-outcome information-equivalence gate. The minimum gate is 12. One candidate is excluded because the independent verifier returns an unparsable audit response; this support failure carries no scientific verdict. All 23 eligible pairs enter downstream execution. Each receives six paired seeds under each memory condition, producing 276 completed requests with zero support failures.

The endpoint asks for the next click from the frozen browser state. The released successful trajectory provides the reference click. We use this endpoint only to localize action-selection divergence under the L1 writer-mode contrast before terminal feedback. Reference-action match is not treated as a surrogate for task success: a different click can still lead to a correct trajectory, so terminal quality remains the distinct endpoint above. Statistical inference uses tasks as the independent units. The preregistered test sign-flips each task-level success-derived minus failure-derived reference-match difference over 100,000 permutations.

For the early-action endpoint, the observed success-derived minus failure-derived reference-action match difference is -0.0942. Its support-side value is 0.9841. The counterevidence-side value is 0.0792. Thus R6 misses both frozen directional criteria: $|\Delta| = 0.0942 < 0.15$ and the relevant randomization value exceeds 0.05. Paired action choices differ in 21.0% of seeds; hypothesized-direction and reverse error-consistent divergence are 0.7% and 10.1%. These are mechanism-localization quantities, not terminal-quality evidence.

4.3 Cross-endpoint adjudication

Table 4 shows opposite descriptive signs on distinct estimands, so we do not pool them. R4 is non-decisive at terminal success; R6 fails both directional criteria on an early-action readout that is not terminal quality. Neither can overwrite the other. The strongest joint statement is *no endpoint-robust directional conclusion under the current L1 writer-mode bundle*; C4 remains active and unrefuted, with no supported provenance-only causal sign.

L3 transport. A source-faithful test requires the motivating writer/checkpoint, original memory interface, fixed acquisition content, and paired source runtime; WebArena is not a substitute. Appendix A.10 gives the execution/decision contract. L3 remains replication debt.

C5 remains a design implication. Provenance is an upstream variable in the memory writer, and the two controlled writer-mode conditions exhibit paired behavioral differences under matched query state. These experiments do not assign those differences a supported provenance-only causal sign, and a downstream performance benefit from provenance-aware governance remains outside the supported claim set.

5 Related Work

Content- and utility-centric agent memory. A-MEM builds dynamically linked agentic memories (Xu et al., 2025), while MemRL learns runtime utility over episodic memories (Zhang et al., 2026). ReasoningBank, Agent Workflow Memory, and Reflexion likewise turn trajectories or feedback into persistent reusable state (Ouyang et al., 2026; Wang et al., 2024; Shinn et al., 2023). These systems establish memory construction and retrieval. Our audit asks which history-dependent channel is being manipulated when future behavior is attributed to provenance.

Provenance, outcome evidence, and trust. Execution-provenance and evidence-tracing work treats lineage and provenance-bearing memory as trust objects (Wang et al., 2026). Memory Provenance Laundering shows that consolidation can hide low-trust source provenance while preserving action triggers and proposes a provenance firewall (Xu et al., 2026). MutMem records signed positive/negative outcome evidence and provenance-bearing authorized memory mutations (Saidi, 2026). These works establish provenance and outcome evidence as memory-governance objects. Our closest-work boundary is narrower: success-derived versus failure-derived *process* provenance, while explicitly separating an outcome-conditioned writer-mode intervention (L1) from a byte-identical provenance-metadata intervention (L2).

Provenance-grounded memory architectures. Eywa separates immutable source evidence from derived beliefs and validates extracted memories against typed source support (Joshi, 2026); in our ladder, that architecture makes L0 auditing natural and could support an L2 audit only if a visible provenance field were varied while the actionable fact stayed fixed. TierMem links lossy summaries to immutable raw pages and escalates retrieval when summary evidence is insufficient (Zhu et al., 2026); its provenance pointers are especially useful for L3 source recovery, while changing the summary/raw route changes available evidence and is therefore not itself an L2 metadata-only intervention. MemORAI tracks turn-level origins in a provenance-enriched graph and uses those relations during adaptive retrieval (Van et al., 2026); toggling such graph edges can change retrieval content, so the corresponding causal object belongs to a retrieval-channel audit unless returned content is held fixed. These examples show the practical role of the ladder: provenance-aware systems already expose governance primitives, but the rung is determined by what the intervention actually changes.

Learning from failure and memory quality. Failure-derived information can be beneficial. Learning from Failure converts failed computer-use trajectories into inference-time diagnoses and patches (Sun et al., 2026); Rethinking Self-Evolving Agent Skills reports useful evolved skills in feedback conditions containing failures (Liu et al., 2026). Experiential Reflective Learning directly separates success- and failure-derived heuristics and reports task-dependent advantages, including stronger failure-derived guidance on its Search split (Allard et al., 2026). Memory Reward Inflation separately shows how erroneous memories can be overvalued and propagated (Asadolahi et al., 2026). Together these results prevent a blanket “failure memory is bad” interpretation and motivate an audit with no assumed causal sign.

Causal identification and claim boundary. The financial-agent audit reports a strong success/failure provenance association (Li & Zhu, 2026), but task difficulty can confound that gap. Existing memory ablations can jointly change information, generation process, retrieval, and metadata. Our L0–L3 protocol requires the intervention channel, hold-fixed variables, independent inference unit, and strongest allowed claim to be declared separately. R4 and R6 are L1 worked examples: they hold future query/state fixed and use a preregistered operational semantic-equivalence rule, but generated memories need not be byte-identical. R5 is the cleaner L2 design because actionable memory is fixed exactly and only provenance metadata changes; its frozen cohort is support-insufficient and executes zero model calls. The contribution is therefore the identification object and resulting unresolved boundary, not a directional causal theorem or a validated provenance-governance policy.

6 LIMITATIONS

The financial AgentDojo audit remains the source of the 0.931-versus-0.647 provenance association and the ten persistent W→W cases. Its public release does not expose the per-query ReasoningBank artifacts required for an exact matched swap. A reconstruction preflight recovered the benchmark and ReasoningBank implementation, but the source configuration uses Qwen 3.7 Flash with qwen3.7-text-embedding and the current environment lacks auditable access to that configuration. Source-faithful L3 replication therefore remains support and experiment debt, not a scientific negative.

L1 is bounded by independent task count and imperfect information matching. R4 has four tasks, $\Delta = +0.1667$, $p = 0.0785$, and minimum exact one-sided $p = 1/16$; its archived cosines 0.902–

0.931 retain all 4/4 pairs at 0.85 and 0.90. Yet an earlier post-hoc stronger-verifier audit judges 0/4 and 1/4 pairs fully equivalent under two reviewers, respectively. Deterministic writer diagnostics likewise find failure-derived memories more caution-heavy in 23/24 R6 pairs, more verification-heavy in 20/24, and more directive in 20/24. R6 itself has 23 disjoint tasks, $\Delta = -0.0942 < 0.15$, and $p_F = 0.0792$ on a non-terminal readout. Both causal gates remain closed; no threshold or sample-size rescue is allowed.

Operational semantic equivalence is weaker than exact information identity, so L1 identifies at most the writer-mode bundle. The 0.15 floor is practical relevance, not a power guarantee; prospective confirmatory planning is specified in Appendix A.9. L2 is stricter because actionable memory is byte-identical, but its frozen ten-task target has only five eligible unique tasks. R5 therefore stops at zero model calls; lowering the target post hoc would violate the design. Instantiating a convenient metadata tag in a different memory architecture could test whether a provenance-label effect exists on that new substrate, but it would not identify the frozen ReasoningBank/financial-agent object. We therefore record such a study as a separate-substrate extension rather than using it to rescue L2 here.

The financial accuracies are rounded; our integer counts are the unique values consistent with released supports, while transition allocation beyond the ten W→W cases is derived from aggregates. WebArena is a controlled bridge, not source-faithful financial AgentDojo transport; L3 remains unavailable (Appendix A.10). Provenance-aware governance has not been shown to improve downstream performance. These boundaries keep U1 and U2 unsupported.

7 Conclusion

The paper's primary methodological output is a provenance-audit identification ladder. L0 asks whether future outcomes differ across observed provenance strata. L1 changes an outcome-conditioned writer under fixed future state and an operational equivalence rule. L2 holds actionable memory byte-identical and changes only visible provenance metadata. L3 tests whether an identified contrast transports to the motivating source runtime. The ladder makes the intervention channel, independent inference unit, identifying assumptions, and strongest allowed claim explicit.

Our evidence populates these rungs without manufacturing a causal sign. The released financial audit provides a strong L0 signal: 0.931 versus 0.647 accuracy and all ten reported persistent W→W failures in the failure-derived stratum. R4 and R6 instantiate L1 but leave both frozen causal gates closed; R4 is structurally non-decisive with four units, while R6 uses a non-terminal action-localization endpoint and fails both the frozen magnitude and significance criteria. Their opposite descriptive signs across different estimands do not support an endpoint-robust direction. R5 targets L2 but support-stops at five eligible unique tasks for a frozen target of ten. Source-faithful L3 replication remains debt. C4 therefore remains active and unrefuted rather than supported or rejected.

This changes what a memory audit should report. A retrieved text is not a complete audit object when the process that produced it is discarded, but provenance is also not a single causal treatment until its channel is specified. Reporting the ladder rung prevents observational association, writer-mode effects, provenance-metadata effects, and transport evidence from being collapsed into one ambiguous "provenance effect." Any policy that converts provenance into trust or reuse decisions still requires separate downstream validation.

Reproducibility Statement

The appendix records the frozen intervention rules and statistical gates, together with the support-failure boundaries used by the controlled experiments. The submission supplement contains the claim ledger and its content-addressed evidence manifest. It also packages experiment receipts with the source needed to rebuild the manuscript and recompute the reported aggregate statistics.

AI Use Disclosure

LLM tools assisted literature triage, methodology implementation, result interpretation, manuscript editing, and mock-review analysis. Scientific claims and reported numbers were checked against frozen artifacts and deterministic scripts; AI judgments were not used as task-success ground truth. The authors reviewed the AI-assisted work and take responsibility for the final content.

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A SOURCE-GROUNDED REANALYSIS AND INTERVENTION DETAILS

A.1 COUNT RECONSTRUCTION

The financial audit reports 101 success-derived retrievals at 0.931 accuracy and 34 failure-derived retrievals at 0.647. The unique integer numerators consistent with three-decimal rounding are 94 and 22, yielding 7 and 12 failures. It also reports ten W→W persistent failures, all failure-derived; hence 10/12 failure-derived errors are persistent, while the success-derived stratum contains no W→W case.

A.2 TERMINAL INFORMATION-EQUIVALENCE CONTRACT

Six terminal candidates are frozen before outcomes. Both memories use the same writer model; eligibility requires valid ReasoningBank items, paired embedding cosine at least 0.75, an independent verifier judging actionable guidance equivalent with no unique task-solving fact, and no literal benchmark-reference leakage. Four candidates pass: tasks $\{387, 167, 23, 388\}$ with cosines 0.931, 0.902, 0.906, and 0.904. The same four remain eligible if the cosine floor is raised from 0.75 to 0.85 or 0.90; this is a zero-call re-thresholding of archived pre-outcome values, not an alternative-verifier run. Task 385 is excluded for repeating "Cally"; task 164 repeats "Dry" and "Uneven color." All exclusions are pre-outcome.

A.3 POST-HOC OPERATIONAL-EQUIVALENCE STRESS TEST

The original information-equivalence gate remains the preregistered R4 admission rule; the following diagnostics do not retroactively change eligibility. An archived stronger-verifier audit judges none of the four admitted R4 pairs fully equivalent under one semantic reviewer and only one of four fully equivalent under a second, with the residual disagreements involving concrete strategy emphasis. A deterministic lexical audit points in the same direction without a new model call: failure-derived memory contains more search/filter language and more strong-directive language in all 6/6 R4 writer pairs; among the 24 R6 writer pairs it contains more failure/caution language in 23/24, more verification language in 20/24, and more strong-directive language in 20/24. These diagnostics make the limitation operational: the success/failure ReasoningBank intervention changes writer mode together with pragmatic register and strategy emphasis. They therefore weaken a provenance-only interpretation of L1 and increase the value of the byte-identical L2 intervention; they do not change either frozen L1 outcome.

A.4 TERMINAL RUNTIME REPAIR BOUNDARY

A condition-blind projection removes repeated global menu blocks, blank lines, and exact duplicate lines without consulting references or outcomes. An initial context-allocation support failure is discarded before scientific admission; the clean restart changes only context capacity. Its largest prompt uses 5,209 of 12,288 input tokens; executor, temperature, seed schedule, pair set, scorer, effect floor, permutation seed, and significance rule remain frozen.

A.5 TERMINAL STATISTICAL RULE

Each qualified pair receives six paired seeds. The statistic is task-level terminal success, success-derived minus failure-derived. Support requires $\Delta \geq 0.15$ and one-sided $p < 0.05$; counterevidence requires $\Delta \leq -0.15$ and its one-sided $p < 0.05$, using 100,000 frozen-seed permutations. Observed $\Delta = +0.1667$ with $p_S = 0.0785$. Four independent task units imply only 16 exact sign assignments and a minimum one-sided value $1/16 = 0.0625$, so this endpoint cannot satisfy the frozen 0.05 rule.

A.6 EARLY-ACTION COHORT AND L1 WRITER-MODE CONTRAST

The disjoint WebArena cohort excludes prior D2 task IDs. A candidate must come from a released successful trajectory and expose a visible single-click reference action at step two or later; eligible tasks are sorted by ID and the first 24 are frozen before counterfactual writing. A compact process trace preserves instruction and action sequence but removes terminal success. The same trace is passed through first-party ReasoningBank success- and failure-reflection prompts using Qwen2.5-32B at temperature zero, directly varying outcome-conditioned writer mode while holding the trace fixed.

Eligibility again requires valid memory items, cosine at least 0.75, verifier-equivalent actionable guidance, no unique task-solving fact, and no reference-action leakage. Twenty-three pairs pass; candidate 159 is excluded for an unparsable verifier output. All 23 enter downstream execution.

A.7 EARLY-ACTION STATISTICAL RULE

Each task receives six paired seeds per writer condition with identical future query and frozen browser state. The output is one visible click index; the released successful trajectory supplies the reference. Inference is task-level: the permutation test sign-flips each task's success-derived minus failure-derived reference-match difference over 100,000 draws, with the same ± 0.15 magnitude and one-sided 0.05 gates. The 23 tasks yield 276 complete requests and zero support failures; mean difference is -0.0942, with $p_S = 0.9841$ and $p_F = 0.0792$. Paired actions differ in 21.0% of seeds; hypothesis-direction error-consistent divergence is 0.7%, versus 10.1% in reverse.

A.8 Exact-information support-capacity stop and reopen condition

R5 holds the actionable memory core byte-identical and changes only structured provenance metadata. Its frozen contract requires ten unique review tasks, but the released shopping split contains only five; it therefore stops before any model call, with no scientific verdict and no post-audit target reduction. The strongest reopen path is source-faithful financial AgentDojo replication of an L1 writer-mode or L2 exact-information contrast when query-level artifacts or auditable exact-model access become available. Such replication addresses L3 transport, not identification by itself; any new bridge must be frozen before outcomes and cannot reuse R4/R6 for threshold rescue.

A.9 Prospective L1 power-planning contract

The current $|\Delta| \geq 0.15$ rule is a practical-relevance floor, not evidence of adequate power. A future confirmatory L1 run must choose the number of independent task pairs before outcomes by simulation or exact enumeration of the planned task-level sign-flip test. The planning distribution must be frozen from an external pilot or other prespecified source; the target effect is $|\Delta| = 0.15$; the directional randomization threshold remains 0.05; and the target prospective power is at least 80%. Pair construction, task exclusions, seeds per condition, maximum task count, and the stopping rule must be frozen with that calculation. Repeated requests within a task may reduce task-level measurement noise but do not count as additional independent units.

For scale, a normal-approximation planning sensitivity for a paired task-level mean at $|\Delta| = 0.15$ gives the following independent-task counts. These are design aids, not retroactive evidence, because the unknown task-level standard deviation must be fixed prospectively from an external pilot.

Planning SD	one-sided $\alpha = .05$, 80% power	two-sided $\alpha = .05$, 80% power
0.20	11	14
0.30	25	32
0.40	44	56

Thus "four tasks" is not defended as an adequately powered design; even the optimistic SD=0.20 planning case points to a double-digit confirmatory cohort. R4 remains structurally non-decisive, and R6 is judged by its frozen effect and significance gates rather than post-hoc sample extension. We retain the preregistered directional randomization tests as the decision tests. A future confirmatory packet should additionally report a two-sided randomization sensitivity and a paired confidence interval; BCa intervals are descriptive rather than a substitute for the frozen randomization gate, and are not retrofitted onto the four-task R4 endpoint.

A.10 Source-faithful L3 decision contract

An L3 transport test must recover the motivating financial-agent writer/checkpoint and original memory interface, freeze the same acquisition trajectory content and downstream schedule across conditions, and manipulate only the declared provenance channel before any outcome is observed. For an L1 source-native test, success- versus failure-conditioned writer mode may change the generated memory, after which the pre-outcome information-equivalence audit is applied and the paired memories are executed in the original runtime. For an L2 source-native test, actionable memory must remain byte-identical while only visible provenance metadata changes.

The current obstacles separate cleanly. *Infrastructure debt* is the missing query-level ReasoningBank artifacts, auditable access to the reported Qwen 3.7 Flash writer and qwen3.7-text-embedding configuration, source checkpoint/prompt bindings, and provenance-bearing runtime fields needed to replay the original financial tasks. *Conceptual requirements* remain even after those assets exist: the intervention channel must be declared before outcomes; L1 must retain a pre-outcome information-equivalence gate; L2 must keep actionable memory byte-identical; and any WebArena-to-financial transport claim must state its target-population assumptions rather than treating source replication as automatic generalization. A writer-output change plus downstream change supports a source-native writer-mode path; a byte-identical-memory metadata contrast with downstream change supports an L2 metadata path on that substrate; and adequate-support null evidence on both channels is substrate-bounded counterevidence rather than a general no-provenance theorem. No such source-faithful execution is claimed in the present paper.